

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing
Milestone-1 Submission

Group Number: 12

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Question-1: What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Solution:

System being studied:

The project analyzes a mass balance model based on energy balance at the surface of the glacier of the Morteratsch Glacier, Switzerland, Alps. The system is a model of the processes of all meteorological and physical variables to determine glacier mass balance of cumulative change within a given time.

Objective:

The objective of the project is to quantify the variability and uncertainty in glacier mass balance using a surface energy balance model and Monte Carlo simulations, and to translate this melt variability into probabilistic estimates of water level (sea-level) rise rather than a single deterministic value.

Probabilistic problem:

Although the mass balance model of a glacier may be calibrated well, there exist random and systematic uncertainty in the input variables and parameters. The stochastic problem is to find out the propagation of these uncertainties using the energy balance model and the resulting impact on the final cumulative mass balance estimate.

Primary sources of uncertainty:

- Uncertainty in the measurement of meteorological data (air temperature, precipitation, radiation, humidity, pressure, etc.)
- Uncertainties in the systematic calibration of the system (e.g. precipitation correction, turbulent exchange coefficient)
- Modelled structural uncertainty that is a result of simplified modelling of physical glacier processes.

Question-2: Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Solution:

Key random variables:

The key random variables of the project are:

- Air temperature
- Snowfall threshold
- Precipitation
- Ice albedo
- Cloudiness
- Other energy balance parameters

Each of these variables has a direct impact on the amount of energy arriving at the glacier surface and, consequently, on the amount of ice that melts.

How uncertainty is modeled

For every input variable, the paper assigns:

- Systematic uncertainty (long-term bias or offset), denoted by ε_s
- Random uncertainty (short-term fluctuations), denoted by ε_r

Random uncertainties vary between model runs, while systematic uncertainties remain constant within a single run but vary across different runs.

Air Temperature Uncertainty Modeling As an example, air temperature uncertainty is modeled by adding both systematic and random error terms to the measured temperature:

$$T_{\text{sim}}(t) = T_{\text{meas}}(t) + \varepsilon_s + \varepsilon_r(t) \quad (1)$$

where:

- $\varepsilon_s \sim \mathcal{N}(0, 0.3^\circ\text{C})$ represents systematic uncertainty and is sampled once per Monte Carlo run (400-day simulation period).
- $\varepsilon_r(t) \sim \mathcal{N}(0, 0.3^\circ\text{C})$ represents random atmospheric variability and is sampled at every time step (hourly).

Both types of uncertainty are modeled using normal (Gaussian) distributions with zero mean. These uncertainties are added to the measured input values during simulation runs as:

$$X = X_{\text{measured}} + \varepsilon_s + \varepsilon_r \quad (2)$$

Probabilistic assumptions:

- Normal distribution of errors

- Independence of variables
- Dominance of systematic errors
- Non-linear sensitivity
- Probabilities snow threshold
- Temporal correlation

Question-3: Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Solution:

Probabilistic dependencies:

- Mass balance is a dependent variable that is affected by various interdependent meteorological and physical variables.
- Components of the energy fluxes (radiation, turbulent fluxes, accumulation) are conditional on temperature, albedo, cloudiness, and humidity.
- Some of the uncertainties (e.g., intensity of precipitation) are considered independent and others (e.g., cloudiness) exhibit time dependence.

Reasoning mechanism:

- The Monte Carlo simulations are repeated draws on the joint uncertainty distributions of each and every parameter.
- Every run yields a cumulative mass balance value and the collection of the runs is a probability distribution and not a single deterministic estimate.
- This allows inference of probable balance effects of mass and their variability which are probabilistic.

Question-4: Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

Solution:

Model Implementation:

- To account for uncertainty in estimates of glacier melt, the surface energy balance model is periodically run using Monte Carlo techniques over a pre-determined simulation time.
- Climate and glaciological input variables, including cloudiness, precipitation, ice albedo, snowfall threshold, and air temperature, drive the model.
- Model tuning involves the correction of precipitation by setting the parameter of precipitation correction (Pcorr) to reflect the disappearance of snow cover as observed at the AWS site.

Alignment with probabilistic model:

- The Monte Carlo model is a direct application of the uncertainty model to ensure that input parameters are varied in every single run.
- Thousands of runs are stored in order to calculate PDFs, means, and standard deviations of mass balance.
- To have stable estimates of uncertainty a number of Monte Carlo runs of about 5000 are enough.

Assumptions affecting design:

- The uncertainties of inputs are independent.
- Uncertainty can be adequately exalted by normal distributions.
- The model is used on one location of a glacier.
- The effect of space is not explicitly modeled.

The resulting distribution of simulated glacier melt is then translated into probabilistic estimates of water level rise.

Question-5: Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

Solution:

Well-defined components:

- Energy balance formulation
- Determination of important uncertain parameters.
- Propagation of uncertainty using Monte Carlo.
- Representation of model uncertainty by use of PDFs.

Aspects expected to evolve:

- More realistic modeling of correlated uncertainties.
- Point-scale glacier modeling to spatial glacier modeling.
- Better correction of system errors.
- Climate model outputs Integration with climate model outputs

Current State of the model: At the current stage, the probabilistic model has a well-defined and implemented framework for uncertainty propagation, particularly for air temperature, which is a dominant driver of glacier melt. The model incorporates uncertainty by perturbing measured temperature values using both systematic and random error components during Monte Carlo simulations.

The simulated air temperature used in the model is given by:

$$T_{\text{sim}}(t) = T_{\text{meas}}(t) + \varepsilon_s + \varepsilon_r(t)$$

where

$$\varepsilon_s \sim \mathcal{N}(0, 0.3^\circ\text{C})$$

represents systematic uncertainty and remains constant within a simulation run, and

$$\varepsilon_r(t) \sim \mathcal{N}(0, 0.3^\circ\text{C})$$

represents random uncertainty and varies with time.

This formulation is already operational and allows the model to realistically capture both long-term bias and short-term atmospheric variability. Through repeated Monte Carlo runs, these uncertainties are propagated through the surface energy balance model, resulting in probabilistic estimates of glacier melt and associated water level rise.

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Question-6: Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Solution:

Unresolved issues

- Uncertainty assumption between uncertain parameters.
- Single point of glacier space representation (limited spatial representation).
- Huge impact of the systematic errors.
- Inability to extrapolate the findings to other glaciers or to the future.

These unresolved issues directly affect the reliability of projected glacier melt and the associated uncertainty in future water level rise estimates.

Who addresses them next:

- Model developers: enhance modeling of uncertainties and correlations
- Data analysts: improve estimates of parameter uncertainties
- Reviewers/implementers: test model performance under varying conditions of data quality